Gotu AI Agriculture

Where Tech with Intelligence Meets Soil

1 Tomato

1.1 Irrigation and Nutrient Management

- **Water requirements and irrigation methods:** Tomato (*Solanum lycopersicum*) is highly sensitive to water application. Light irrigation is provided 3–4 days after transplanting. Irrigation intervals are scheduled according to soil type and rainfall, specifically 7–8 days interval during kharif, 10–12 days during rabi, and 5–6 days during summer. Critical crop stages requiring continuous water supply without water stress are flowering and fruit development. Severe water stress followed by heavy irrigation causes fruit cracking and must be avoided. To minimize flower and fruit drop during periods of high ambient temperature forecast exceeding 40.0
(°
)$C, sufficient irrigation must be provided. Irrigation methods documented include drip irrigation, u$
$trays used to ensure adequate drainage and prevent damping-off. Quantitative crop evapotranspira$

- **Soil and water conditions:** Tomato cultivation is documented on medium black soils and loamy light soils. Soil conditions such as high soil moisture combined with high humidity, cloudiness, and temperatures below 24
(°
)$C favor damping-off fungal infection, while crowded seedlings, excess soil solutes, and dampnes$
$off and buckeye fruit rot. Blossom-end rot is aggravated by abiotic soil and water conditions including$

- **Nutrient requirements:** Essential macro-nutrients required for tomato growth include nitrogen, phosphorus, and potassium, alongside secondary and micronutrients such as calcium, sulfur, magnesium, zinc, boron, copper, iron, manganese, and molybdenum. Nitrogen promotes vegetative growth, but excessive nitrogen causes over-succulence, increasing susceptibility to insect pests and diseases such as serpentine leaf miner, puffiness, and gray wall, whereas low nitrogen retards plant growth. Phosphatic fertilizers exhibit residual soil availability from previous crop seasons and should not be applied excessively every season. Potassic fertilizers enhance fruit quality and reduce physiological disorders such as gray wall and internal white tissue. Recommended actual nutrient requirements for varieties are 40 kg N, 24 kg
(P_2O_5
), *and* 24 *kg*
(K_2O
)*per acre* (*equivalent to* 40 : 24 : 24 *kg N* : *P* : *K per acre*), *or alternatively documented as* 120 *kg N*, 80
(P_2O_5
), *and* 50 *kg*

(K_2O
)$per hectare. For hybrid tomatoes, nutrient requirements are 60\ kg N, 36\ kg$
(P_2O_5
), $and 36\ kg$
(K_2O
)$per acre (60 : 36 : 36\ kg N : P : K per acre), or alternatively documented as 180\ kg N, 100\ kg$
(P_2O_5
), $and 60\ kg$
(K_2O
)$per hectare. Deficient micronutrients should be identified via soil testing and applied to the soil at sou$
$specific nutrient uptake rates per plant biomass unit are Not specified in the provided sources.$

- **Fertilizers and application methods:** Soil application of organic manures includes incorporating 15–20 tonnes per hectare of well-decomposed farmyard manure (FYM) into the soil, or 8–10 tonnes per acre of well-rotten FYM incorporated 2–3 weeks before transplanting, or 5 tonnes per acre of vermicompost incorporated 1 week before transplanting. Enriched FYM may be applied at 2 tonnes per hectare incorporated with organic formulations of *Pseudomonas fluorescens* and *Trichoderma harzianum* or *Trichoderma asperellum* at 2 kg per tonne of FYM. Neem cake is applied at 100 kg per acre during field preparation. For inorganic fertilizer product application in varieties on a per-acre basis, 50% of the nitrogen dose (20 kg N per acre) and 100% of phosphatic and potassic fertilizers (24 kg
(P_2O_5
)$and 24\ kg$
(K_2O
)$per acre) are applied as a basal dressing before transplanting, followed by a second nitrogen dose of 13$
$dressed at 45 days after transplanting, and a third nitrogen dose of 13.5\ kg N per acre at 60 days after tr$
$hectare basis for varieties, half the nitrogen dose (60\ kg N) and full phosphorus (80\ kg$
(P_2O_5
))$and potassium (50\ kg$
(K_2O
))$are applied at planting, with the remaining 60\ kg N top-dressed 30 days after transplanting. For hy$
$acre basis, nitrogen is split into three equal doses (first at last ploughing, second dose of 20\ kg N per acr$
$hectare basis for hybrids, 60\ kg N and half of P and K (50\ kg$
(P_2O_5
)$and 30\ kg$
(K_2O
))$are given at transplanting, 60\ kg N and the remaining P and K are top-dressed 30 days after transpl$
$dressed 50 days after transplanting. Biofertilizers include seed and seedling treatments with Azotob$
$-10\ g each per kg of seed, seedling root dip with Azotobacter and PSB at 250\ g each per acre of seedlings,$
$-3 days before sowing, or Triacontanol 0.05\% EC / 0.1\% EW sprayed at 100\ ml in 160 -$
$-200\ liters of water per acre at 25, 45, and 65 days after sowing; Gibberellic acid (GA) sprayed at 0.001%$
$-200\ liters of water per acre) at 45 and 65 days after sowing, or GA foliar spray at 10 -$
$-20\ mg per liter for low-temperature yield enhancement and 40 - -100\ mg per liter seed treatment;$
$-500\ mg per liter for flowering / fruiting and 1000\ mg per liter pre-harvest spray for fruit ripening$
$-100\ mg per liter during flowering to enhance fruit set under temperature extremes. Specific comme$

- **Water and nutrient disorders:** Water stress during flowering and fruit set leads

to severe flower and fruit drop and reduced fruit size, while sudden irrigation following water stress causes fruit cracking. Excessive watering or poor drainage induces waterlogging and damping-off disease. Major nutrient deficiency symptoms, causes, and documented corrective measures include: Nitrogen deficiency causes uniform yellowing (chlorosis) of older leaves progressing to yellowish white, smaller top leaves, and spindly growth; corrected by top-dressing urea based on soil tests or foliar spraying 2% urea solution. Phosphorus deficiency produces necrotic spots on older leaves, stunted growth, and purpling of stems, petioles, and leaf undersides; corrected by soil application of recommended phosphatic fertilizers at sowing/planting. Potassium deficiency causes young leaf marginal tip burn, interveinal necrosis, curling, and crinkling with irreversible chlorosis; corrected by foliar spray of 1%
(K_2SO_4
$).Sulfurdeficiencycausesuniformchlorosisacrossyoungandoldleaveswithdistinctreddishcolora$
(K_2SO_4
$)or1\%$
($CaSO_4$
$)twiceatfortnightlyintervals.Magnesiumdeficiencycausesadvanceinterveinalchlorosisstartin$
($MgSO_4$
$).Manganesedeficiencycauseslightinterveinalchlorosisprogressingtodarknecroticareasalongve$
($MnSO_4$
$).Molybdenumdeficiencycausesmottledinterveinalchlorosiswithoutreddishleafundersides;cor$
(Na_2MoO_4
$)twiceatweeklyintervals.Zincdeficiencycausesinterveinalnecrosis,yellowingofyoungleaves,an$
($ZnSO_4$
$).Borondeficiencycauseslightgeneralchlorosis,meristematictissuenecrosis,lossofapicaldomin$
$EndRot(BER)characterizedbytan,water-soaked,blackleatherylesionsattheblossomendoffruit$
$graybleachingwithsunkennecroticspots;correctedbyfoliarsprayof0.5\%$
($CuSO_4$
$)twiceatfortnightlyintervals.Irondeficiencycausesinterveinalchlorosisontheyoungestleaveseve$
($FeSO_4$
$).Physiologicalfruitdisordersinclude Cat-face(grossblossom-enddeformitycausedbylowtempe$
$burstgreen/yellowwalldiscoloredareascausedbysilverleafwhiteflyBemisia\ argentifoliinymphfe$

- **Monitoring and scheduling:** Soil testing must be conducted prior to sowing or transplanting to assess soil fertility status and determine required macronutrient and micronutrient application rates. Agro-ecosystem field monitoring should be conducted regularly at least once a week to evaluate soil conditions, water availability, crop health, weather parameters, and pest-to-defender ratios. Irrigation schedules are determined based on season (7–8 days in kharif, 10–12 days in rabi, 5–6 days in summer). Split fertilizer schedules are executed at specific intervals (basal at planting/ploughing, followed by top dressing at 30, 45, 50, and 60 days after transplanting or planting depending on crop type). Quantitative soil tension values, tensiometer thresholds, sap test diagnostic benchmarks, and formal record-keeping formats are Not specified in the provided sources.

- **Effects on crop performance:** Sufficient irrigation during high temperature forecast periods ($>40.0

(°
)C)$prevents flower and fruit drop, while maintaining consistent soil moisture prevents fruit cracki$
$facing. Combining drip irrigation with plant staking reduces foliage dampness, improves spray cove$
$-55 tonnes per hectare (e.g., 40--45 tonnes per hectare for determinate processing cultivars, 50 ton$
$-55 tonnes per hectare for Phule Raja hybrid), while experimental rainy season hybrid yields range f$
$use efficiency($
(kg/m^3
))$and nutrient - use efficiency indices are Not specified in the provided sources.$

- **Integrated management workflow:** The documented sequence for integrated tomato irrigation and nutrient management proceeds as follows: First, conduct pre-sowing soil testing to assess soil fertility status and identify specific micronutrient deficiencies. Second, prepare the field through deep summer ploughing and incorporate well-decomposed FYM (15–20 tonnes per hectare or 8–10 tonnes per acre) or vermicompost (5 tonnes per acre) or bio-enriched FYM (2 tonnes per hectare) alongside neem cake (100 kg per acre) during field preparation 1–3 weeks prior to transplanting. Third, perform seed and seedling bio-inoculation using *Azotobacter*, phosphorus solubilizing bacteria, *Trichoderma viride*, or *Pseudomonas fluorescens* (saving up to 50% N and 25% P inorganic fertilizer requirements). Fourth, establish field planting on raised beds (15 cm height) or broad bed furrows with drip or surface irrigation systems, applying basal inorganic fertilizers (50% N and 100% P and K for varieties; 1/3 N and 1/2 P and K for hybrids). Fifth, implement weekly field monitoring of crop health, soil moisture, and climatic factors, applying scheduled irrigation at 5–12 day intervals depending on season while ensuring water stress is strictly avoided during flowering and fruit development. Sixth, execute stage-wise split top-dressing of nitrogen and potassium at 30, 45, 50, and 60 days after transplanting or planting. Seventh, diagnose abiotic disorders or nutritional deficiencies during crop growth and execute targeted corrective measures, such as applying 2% urea for nitrogen deficiency, 1%
(K_2SO_4
)$for potassium deficiency, 2\%$
($CaSO_4$
)$for calcium deficiency/BER, or 0.2\% borax for boron deficiency. Eighth, evaluate crop per forman$
$benefit workflow models are Not specified in the provided sources.$